Year 12 Maths - Quadratics

Similar questions - Quadratic inequalities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.**

A farmer uses 40 m of fencing to make three sides of a rectangular enclosure. The fourth side is a wall. The sides perpendicular to the wall have length x m.

(a) Show that the area of the enclosure is $(40x - 2x^2)$ m^2. **[1]**

(b) The area must be at least 150 m^2. By solving a suitable inequality, find the possible values of x. **[4]**

(c) The side parallel to the wall must also be at least 12 m. Find the possible values of x. **[2]**

2 Find the set of values of x for which both $3x^2 - x - 2 > 0$ and $x^2 - 4 < 0$. Give your answer in set notation. **[6]**

Question 1

(a) Side parallel to the wall $= 40 - 2x$, so area $= x(40-2x) = 40x - 2x^2$

(b) $40x - 2x^2 \geq 150 \Rightarrow x^2 - 20x + 75 \leq 0 \Rightarrow (x-5)(x-15) \leq 0$, so $5 \leq x \leq 15$

(c) $40 - 2x \geq 12 \Rightarrow x \leq 14$. Combined with part (b): $5 \leq x \leq 14$

Question 2

$3x^2 - x - 2 = (3x+2)(x-1) > 0 \Rightarrow x < -\frac{2}{3}$ or $x > 1$. $x^2 - 4 < 0 \Rightarrow -2 < x < 2$. Both hold when $\{x : -2 < x < -\frac{2}{3}\} \cup \{x : 1 < x < 2\}$